



Module 4 challenge

1. A data analyst at a recycling company manually recalculates the new column materials_sorter. They want to identify any rows with values that do not match those in the original column, compost_sorter. Which SQL clauses would enable them to do so? Select all that apply.

1 / 1 point

☐ WHERE materials_sorter >< compost_sorter

☒ WHERE materials_sorter <> compost_sorter

✓ Correct

☒ WHERE materials_sorter != compost_sorter

✓ Correct

☐ WHERE materials_sorter !! compost_sorter

2. Fill in the blank: To group table rows with the same values into_____, a data analyst uses the SQL command **GROUP BY**.

1 / 1 point

- ☐ new columns
- ☐ a temporary table
- ☒ summary rows
- ☐ an aggregate table

✓ Correct

3. What will this spreadsheet function return?

1 / 1 point

`=SUMIF(E2:E10, ">=50", F2:F10)`

- ☐ The count of the number of cells in the array E2:E10 that have a value greater than or equal to 50.
- ☐ The sum of all values in cells E2 to E10 for which the value in cells F2 to F10 is greater than or equal to 50.
- ☒ The sum of all values in cells F2 to F10 that correspond to values in cells E2 to E10 that are greater than or equal to 50.
- ☐ The sum of any values in cells E2 to E10 and cells F2 to F10 that are greater than or equal to 50.

✓ Correct

4. Which of the following statements accurately describe pivot tables? Select all that apply.

1 / 1 point

- ☒ The columns of a pivot table organize and display values vertically.

✓ Correct

- ☒ The filters section of a pivot table is used to apply filters based on specific criteria.

✓ Correct

- ☐ The rows of a pivot table organize and display values vertically.

- ☒ A pivot table can be used to sort, reorganize, or group data.

✓ Correct

5. A data analyst at a party planning business reviews attendee counts. They calculate the number of spreadsheet rows that contain values less than 500. Which function do they use?

1 / 1 point

- ☐ `=COUNTIF(L10:L300, ">500")`
- ☐ `=COUNTIF("<500", L10:L300)`
- ☐ `=COUNTIF(">500", L10:L300)`
- ☒ `=COUNTIF(L10:L300, "<500")`

✓ Correct

6. Besides potentially improving query performance or efficiency, what is another significant advantage of using structures like temporary tables or **WITH** clauses when writing SQL queries?

1 / 1 point

- ☐ They automatically add comments to the SQL code explaining each step.
- ☐ They guarantee that the results stored within them are automatically indexed for faster retrieval.
- ☐ They eliminate the need to use **WHERE** clauses for filtering data.
- ☒ They help make complex SQL queries more organized, readable, and easier for team members to understand.

✓ Correct

7. Which SQL statement will create a temporary table?

1 / 1 point

☐

```
1 CREATE TABLE temp_table AS (  
2     SELECT orig_table  
3 );
```

☒

```
1 WITH temp_table AS (  
2     SELECT *  
3     FROM orig_table WHERE y = 1  
4 );
```

☐

```
1 SELECT *  
2 FROM temp_table;
```

☐

```
1 WITH temp_table FROM (  
2     SELECT *  
3     = orig_table  
4 );
```

✓ Correct

8. Which column of the original data was set as a "Row" when making this pivot table?

1 / 1 point

day	AVERAGE of tip	AVERAGE of total_bill
Thursday	\$2.77	\$17.68
Friday	\$2.73	\$17.15
Saturday	\$2.99	\$20.44
Sunday	\$3.26	\$21.41

- ☐ AVERAGE
- ☐ total_bill
- ☐ tip
- ☒ day

✓ Correct